

KAD Power — Comprehensive UI/UX Color System

Advanced Design Token Architecture | Light & Dark Modes

Version: 1.0.0

Based on: KAD Power Visual Identity System (Sections 3, 4, 5)

Scope: Web Application Components, Typography, Surfaces, Interactions, Data Visualization, Accessibility

Philosophy: 60% Neutral / 30% Navy Structure / 10% Green Energy

1. Design Principles

- Structural Hierarchy:** Navy provides depth, trust, and structural integrity.
- Energy Accent:** Solar Green is reserved for emphasis, activity, success, and positive contrast (10% rule).
- Neutral Foundation:** Slate neutrals carry 60% of the UI surface to ensure readability and reduce eye strain.
- Contrast First:** Every token pair is validated against WCAG 2.1 AA standards (minimum 4.5:1 for text, 3:1 for UI components).
- Mode Integrity:** Dark mode is not an inversion; it is a re-orchestration of depth where Navy becomes the atmospheric background and Green retains its luminance as the active signal.

2. Primitive Color Tokens (Core Palette)

These are the immutable raw colors. Do not use them directly in components; reference them via Semantic Tokens.

2.1 Navy — Structural Family

Token	Hex	Usage Context
navy-50	#F4F4F8	Lightest tint, subtle backgrounds
navy-100	#E4E6ED	Dividers, disabled states
navy-200	#C3C8D9	Borders on light surfaces
navy-300	#9FA6C1	Secondary icons, placeholder text
navy-400	#5F6B97	Muted structural elements
navy-500	#1A2B6B	Brand Primary — buttons, links, headers
navy-600	#18275F	Hover states for navy-500
navy-700	#162251	Active/pressed states
navy-800	#131E43	Deep surfaces, dark mode cards
navy-900	#111935	Deepest background, darkest text on light mode

2.2 Solar Green — Energy Family

Token	Hex	Usage Context
green-50	#FAFEF8	Subtle success backgrounds
green-100	#F3FEED	Success tint surfaces
green-200	#E4FCD9	Hover on green tint backgrounds
green-300	#D4FAC1	Light emphasis, glow gradients
green-400	#B8F798	Highlights, progress fills
green-500	#99F36C	Brand Accent — primary actions, active states, success
green-600	#85D360	Hover on green-500
green-700	#6EAE52	Active/pressed accent
green-800	#588A44	Dark mode success text
green-900	#416535	Deepest green, dark mode success bg

2.3 Slate — Neutral Family (Navy-Tempered)

Token	Hex	Usage Context
slate-0	#FFFFFF	Pure white, highest elevation
slate-50	#F6F7F9	Primary light background
slate-100	#EAEBF0	Secondary background, card hover
slate-200	#D6D8E1	Borders, dividers
slate-300	#B8BCCC	Disabled text, inactive icons
slate-400	#8F95AE	Tertiary text, placeholders
slate-500	#6B7494	Secondary text, captions
slate-600	#565D76	Medium emphasis text
slate-700	#404559	Strong secondary text
slate-800	#2D313E	Deepest neutral, dark mode secondary text

2.4 Extended Functional Palette (UI Necessities)

Derived to maintain system harmony while serving critical UI states (error, warning, info). These are **functional extensions**, not brand identity colors.

Token	Hex	Family	Usage
func-error-50	#FEF2F2	Error tint	Error background
func-error-100	#FEE2E2	Error light	Error border
func-error-500	#EF4444	Error core	Error text, icons, destructive actions
func-error-700	#B91C1C	Error dark	Dark mode error
func-warn-50	#FFFBE8	Warning tint	Warning background
func-warn-100	#FEF3C7	Warning light	Warning border
func-warn-500	#F59E0B	Warning core	Warning text, icons
func-warn-700	#B45309	Warning dark	Dark mode warning
func-info-50	#EFF6FF	Info tint	Info background
func-info-100	#DBEAFE	Info light	Info border
func-info-500	#3B82F6	Info core	Info text, icons
func-info-700	#1D4ED8	Info dark	Dark mode info

3. Semantic Color Tokens

Semantic tokens map primitives to meaning (background, text, border). They have dual values: **Light** and **Dark**.

3.1 Background Tokens

Token	Light Mode		Dark Mode		CSS Variable
bg-page	slate-50	#F6F7F9	navy-900	#111935	--bg-page
bg-surface	slate-0	#FFFFFF	navy-800	#131E43	--bg-surface
bg-surface-elevated	slate-0	#FFFFFF	navy-700	#162251	--bg-surface-elevated
bg-surface-overlay	slate-0	#FFFFFF	navy-800	#131E43	--bg-surface-overlay
bg-primary	navy-500	#1A2B6B	navy-400	#5F6B97	--bg-primary
bg-primary-hover	navy-600	#18275F	navy-300	#9FA6C1	--bg-primary-hover
bg-primary-active	navy-700	#162251	navy-200	#C3C8D9	--bg-primary-active
bg-accent	green-500	#99F36C	green-500	#99F36C	--bg-accent
bg-accent-hover	green-600	#85D360	green-400	#B8F798	--bg-accent-hover
bg-accent-active	green-700	#6EAE52	green-300	#D4FAC1	--bg-accent-active
bg-muted	navy-50	#F4F4F8	navy-800	#131E43	--bg-muted
bg-success	green-50	#FAFEF8	green-900	#416535	--bg-success
bg-warning	func-warn-50	#FFFBE8	func-warn-700	#B45309	--bg-warning
bg-error	func-error-50	#FEF2F2	func-error-700	#B91C1C	--bg-error
bg-info	func-info-50	#EFF6FF	func-info-700	#1D4ED8	--bg-info

3.2 Text Tokens

Token	Light Mode	Dark Mode	CSS Variable
text-primary	navy-900 #111935	slate-0 #FFFFFF	--text-primary
text-secondary	slate-500 #6B7494	slate-300 #B8BCCC	--text-secondary
text-tertiary	slate-400 #8F95AE	slate-500 #6B7494	--text-tertiary
text-placeholder	slate-300 #B8BCCC	slate-600 #565D76	--text-placeholder
text-inverse	slate-0 #FFFFFF	navy-900 #111935	--text-inverse
text-on-primary	slate-0 #FFFFFF	navy-900 #111935	--text-on-primary
text-on-accent	navy-900 #111935	navy-900 #111935	--text-on-accent
text-link	navy-500 #1A2B6B	green-400 #B8F798	--text-link
text-link-hover	green-600 #85D360	green-300 #D4FAC1	--text-link-hover
text-success	green-700 #6AEAE52	green-400 #B8F798	--text-success
text-warning	func-warn-700 #B45309	func-warn-100 #FEF3C7	--text-warning
text-error	func-error-700 #B91C1C	func-error-100 #FEE2E2	--text-error
text-info	func-info-700 #1D4ED8	func-info-100 #DBEAFE	--text-info

3.3 Border Tokens

Token	Light Mode	Dark Mode	CSS Variable
border-default	slate-200 #D6D8E1	navy-700 #162251	--border-default
border-subtle	slate-100 #EAEBF0	navy-800 #131E43	--border-subtle
border-strong	slate-300 #B8BCCC	navy-600 #18275F	--border-strong
border-primary	navy-500 #1A2B6B	navy-400 #5F6B97	--border-primary
border-accent	green-500 #99F36C	green-500 #99F36C	--border-accent
border-focus	green-500 #99F36C	green-400 #B8F798	--border-focus
border-error	func-error-500 #EF4444	func-error-500 #EF4444	--border-error
border-success	green-600 #85D360	green-600 #85D360	--border-success

3.4 Icon Tokens

Token	Light Mode	Dark Mode	CSS Variable
icon-primary	navy-900 #111935	slate-0 #FFFFFF	--icon-primary
icon-secondary	slate-500 #6B7494	slate-300 #B8BCCC	--icon-secondary
icon-tertiary	slate-400 #8F95AE	slate-500 #6B7494	--icon-tertiary
icon-on-primary	slate-0 #FFFFFF	navy-900 #111935	--icon-on-primary
icon-on-accent	navy-900 #111935	navy-900 #111935	--icon-on-accent
icon-accent	green-500 #99F36C	green-500 #99F36C	--icon-accent
icon-success	green-600 #85D360	green-400 #B8F798	--icon-success
icon-warning	func-warn-500 #F59E0B	func-warn-500 #F59E0B	--icon-warning
icon-error	func-error-500 #EF4444	func-error-500 #EF4444	--icon-error

4. Component-Level Color Tokens

4.1 Buttons

Primary Button (Navy)

State	Background	Text	Border	Shadow
Default	bg-primary — navy-500	text-on-primary — White	transparent	none
Hover	bg-primary-hover — navy-600	text-on-primary — White	transparent	shadow-sm
Active	bg-primary-active — navy-700	text-on-primary — White	transparent	inset shadow
Disabled	slate-100	slate-300	transparent	none
Focus Ring	transparent	—	border-focus — green-500	0 0 0 3px green-500/30

Secondary Button (Outline)

State	Background	Text	Border
Default	transparent	navy-500	navy-500 1.5px
Hover	navy-50	navy-600	navy-600
Active	navy-100	navy-700	navy-700
Disabled	transparent	slate-300	slate-200
Focus Ring	—	—	green-500 0 0 0 3px green-500/30

Accent Button (Green)

State	Background	Text	Border
Default	green-500	navy-900	transparent
Hover	green-600	navy-900	transparent
Active	green-700	navy-900	transparent
Disabled	slate-100	slate-300	transparent
Focus Ring	—	—	green-500 0 0 0 3px green-500/30

Ghost Button

State	Background	Text
Default	transparent	navy-500
Hover	navy-50	navy-600
Active	navy-100	navy-700
Disabled	transparent	slate-300

Destructive Button

State	Background	Text
Default	func-error-500	White
Hover	func-error-700	White
Active	#991B1B	White

4.2 Form Inputs & Textareas

Element	Light Mode	Dark Mode
Background	slate-0 White	navy-800 #131E43
Text	text-primary navy-900	text-primary White
Placeholder	text-placeholder slate-300	text-placeholder slate-600
Border Default	border-default slate-200	border-default navy-700
Border Hover	slate-300	navy-600
Border Focus	border-focus green-500	green-400
Border Error	border-error func-error-500	func-error-500
Border Success	border-success green-600	green-600
Shadow Focus	0 0 0 3px rgba(153,243,108,0.25)	0 0 0 3px rgba(153,243,108,0.20)
Icon	icon-secondary slate-500	icon-secondary slate-300
Label	text-primary navy-900	text-primary White
Helper Text	text-secondary slate-500	text-secondary slate-300
Error Text	text-error func-error-700	text-error func-error-100

4.3 Cards & Containers

Variant	Background	Border	Shadow (Light)	Shadow (Dark)
Default Card	bg-surface White	border-subtle slate-100	0 1px 3px rgba(26,43,107,0.08)	0 1px 3px rgba(0,0,0,0.30)
Elevated Card	bg-surface White	border-subtle	0 4px 12px rgba(26,43,107,0.12)	0 4px 12px rgba(0,0,0,0.40)
Outlined Card	bg-surface White	border-default slate-200	none	none
Interactive Card Hover	bg-surface	border-primary navy-500	0 4px 16px rgba(26,43,107,0.15)	0 4px 16px rgba(0,0,0,0.50)
Muted Card	bg-muted navy-50	transparent	none	none
Dark Card (always dark)	navy-800	navy-700	0 2px 8px rgba(0,0,0,0.40)	—

4.4 Navigation (Header / Sidebar)

Header (Top Nav)

Element	Light Mode	Dark Mode
Background	bg-surface White	navy-900 #111935
Border Bottom	border-subtle slate-100	navy-800 #131E43
Logo Text	navy-900	green-500
Nav Link Default	text-secondary slate-500	slate-300
Nav Link Hover	navy-500	green-400
Nav Link Active	navy-900 + green-500 underline	White + green-500 underline
Nav Link Active BG	navy-50	navy-800
Icon Default	icon-secondary	slate-300
Icon Hover	navy-500	green-400

Sidebar

Element	Light Mode	Dark Mode
Background	slate-0 White	navy-900 #111935
Border Right	border-subtle	navy-800
Item Default	text-secondary	slate-300
Item Hover	navy-50 bg, navy-600 text	navy-800 bg, green-400 text
Item Active	navy-500 bg, White text	navy-700 bg, green-500 text
Item Active Border Left	green-500 3px	green-500 3px
Submenu Background	slate-50	navy-800
Collapse Icon	slate-400	slate-500

4.5 Modal / Dialog / Drawer

Element	Light Mode	Dark Mode
Overlay Backdrop	navy-900 / 60% opacity	black / 70% opacity
Modal Background	bg-surface White	navy-800 #131E43
Modal Border	border-subtle	navy-700
Header Background	slate-50	navy-900
Header Title	text-primary	text-primary
Header Border	border-subtle	navy-700
Close Icon	slate-400	slate-400
Close Icon Hover	func-error-500	func-error-500
Footer Background	slate-50	navy-900
Footer Border	border-subtle	navy-700

4.6 Tables

Element	Light Mode	Dark Mode
Header Background	navy-50	navy-800
Header Text	navy-900	slate-0
Header Border Bottom	navy-200	navy-700
Row Default	bg-surface White	navy-800
Row Alternate	slate-50	navy-900
Row Hover	navy-50	navy-700
Row Selected	green-50	green-900 / 40% opacity
Cell Text	text-primary	text-primary
Cell Border	border-subtle	navy-700
Sort Icon Active	green-500	green-500
Pagination Background	slate-0	navy-900
Pagination Active	navy-500 bg, White text	green-500 bg, navy-900 text

4.7 Tabs

Variant / State	Light Mode	Dark Mode
Underline Tabs		
Tab Default	text-secondary	slate-300
Tab Hover	navy-600 text	green-400 text
Tab Active	navy-900 text + green-500 underline 2px	White text + green-500 underline 2px
Tab Disabled	slate-300 text	slate-600 text
Pill Tabs		
Tab Default	transparent bg, text-secondary	transparent bg, slate-300
Tab Hover	navy-50 bg	navy-800 bg
Tab Active	navy-500 bg, White text	green-500 bg, navy-900 text
Contained Tabs		
Container BG	slate-100	navy-800
Tab Default	transparent	transparent
Tab Active	bg-surface White + shadow	navy-700 bg + green-500 top border

4.8 Alerts & Banners

Type	Background	Border	Icon	Text
Success	green-50	green-200	green-600	green-800
Success (Dark)	green-900 / 30%	green-700	green-400	green-100
Warning	func-warn-50	func-warn-100	func-warn-500	func-warn-700
Warning (Dark)	func-warn-700 / 30%	func-warn-500	func-warn-500	func-warn-100
Error	func-error-50	func-error-100	func-error-500	func-error-700
Error (Dark)	func-error-700 / 30%	func-error-500	func-error-500	func-error-100
Info	func-info-50	func-info-100	func-info-500	func-info-700
Info (Dark)	func-info-700 / 30%	func-info-500	func-info-500	func-info-100
Neutral	slate-50	slate-200	slate-500	slate-700
Neutral (Dark)	navy-800	navy-700	slate-400	slate-200

4.9 Badges & Tags (Chips)

Variant	Background	Text	Border
Default	slate-100	slate-700	transparent
Primary	navy-500	White	transparent
Accent	green-500	navy-900	transparent
Success	green-100	green-800	green-200
Warning	func-warn-100	func-warn-700	func-warn-200
Error	func-error-100	func-error-700	func-error-200
Info	func-info-100	func-info-700	func-info-200
Outline	transparent	navy-500	navy-500
Ghost	transparent	slate-500	slate-200

Dark Mode Adjustments: - Default: navy-800 bg, slate-200 text - Primary: navy-400 bg, navy-900 text - Accent: green-500 bg, navy-900 text (unchanged) - Success: green-900 / 50% bg, green-300 text - Warning: func-warn-700 / 30% bg, func-warn-100 text - Error: func-error-700 / 30% bg, func-error-100 text - Info: func-info-700 / 30% bg, func-info-100 text

4.10 Progress & Loading

Element	Light Mode	Dark Mode
Track Background	slate-200	navy-700
Progress Fill (Default)	navy-500	navy-400
Progress Fill (Success)	green-500	green-500
Progress Fill (Warning)	func-warn-500	func-warn-500
Progress Fill (Error)	func-error-500	func-error-500
Spinner Primary	navy-500	navy-400
Spinner Accent	green-500	green-500
Skeleton Base	slate-100	navy-800
Skeleton Shimmer	slate-50 → slate-200 gradient	navy-700 → navy-600 gradient

4.11 Stepper / Timeline / Breadcrumbs

Element	Light Mode	Dark Mode
Step Complete	green-500 bg, navy-900 icon	green-500 bg, navy-900 icon
Step Active	navy-500 bg, White icon	green-500 bg, navy-900 icon
Step Pending	slate-100 bg, slate-400 icon	navy-800 bg, slate-500 icon
Connector Complete	green-500	green-500
Connector Pending	slate-200	navy-700
Breadcrumb Default	text-secondary	slate-300
Breadcrumb Hover	navy-500 + underline	green-400 + underline
Breadcrumb Active	text-primary	text-primary
Breadcrumb Separator	slate-300	navy-600

4.12 Tooltip & Popover

Element	Light Mode	Dark Mode
Background	navy-900	slate-0 White
Text	White	navy-900
Border	transparent	slate-200
Shadow	0 4px 12px rgba(17,25,53,0.25)	0 4px 12px rgba(0,0,0,0.20)
Arrow	navy-900	slate-0

4.13 Toggle Switch

State	Light Mode	Dark Mode
Track Off	slate-200	navy-700
Track On	green-500	green-500
Thumb	White	slate-0
Thumb Shadow	0 1px 3px rgba(0,0,0,0.20)	0 1px 3px rgba(0,0,0,0.40)
Focus Ring	0 0 0 3px green-500/30	0 0 0 3px green-500/30

4.14 Checkbox & Radio

State	Light Mode	Dark Mode
Box Default	White bg, slate-300 border	navy-800 bg, navy-600 border
Box Hover	navy-50 bg, navy-400 border	navy-700 bg, navy-500 border
Box Checked	navy-500 bg, navy-500 border	green-500 bg, green-500 border
Checkmark	White	navy-900
Indeterminate	navy-500 bg, White dash	green-500 bg, navy-900 dash
Disabled	slate-100 bg, slate-200 border	navy-800 bg, navy-700 border
Focus Ring	0 0 0 3px green-500/30	0 0 0 3px green-500/30

4.15 Slider / Range

Element	Light Mode	Dark Mode
Track	slate-200	navy-700
Active Track	navy-500	green-500
Thumb	White, slate-300 border	navy-800 , navy-500 border
Thumb Hover	green-50 shadow	green-500 glow
Thumb Active	green-500 border	green-400 border
Mark Default	slate-300	navy-600
Mark Active	navy-500	green-500

4.16 Dropdown / Select / Autocomplete

Element	Light Mode	Dark Mode
Trigger Background	White	navy-800
Trigger Border	slate-200	navy-700
Menu Background	White	navy-800
Menu Border	slate-200	navy-700
Menu Shadow	0 8px 24px rgba(26,43,107,0.12)	0 8px 24px rgba(0,0,0,0.40)
Item Default	text-primary	text-primary
Item Hover	navy-50 bg	navy-700 bg
Item Active/Selected	green-50 bg, navy-900 text	green-900 /30% bg, green-300 text
Item Icon	icon-secondary	icon-secondary
Group Header	slate-500 text, slate-50 bg	slate-400 text, navy-900 bg
Divider	slate-100	navy-700

4.17 Accordion / Collapsible

Element	Light Mode	Dark Mode
Container Border	border-subtle	navy-700
Header Default	text-primary	text-primary
Header Hover	navy-50 bg	navy-800 bg
Header Expanded	navy-500 text + green-500 left border	green-400 text + green-500 left border
Chevron Icon	slate-400	slate-500
Chevron Expanded	green-500	green-500
Content Background	slate-50	navy-900
Content Text	text-secondary	text-secondary

4.18 Toast / Snackbar

Type	Background	Text	Icon	Border / Accent
Default	navy-900	White	slate-0	Left: slate-400
Success	navy-900	White	green-500	Left: green-500
Error	navy-900	White	func-error-500	Left: func-error-500
Warning	navy-900	White	func-warn-500	Left: func-warn-500
Info	navy-900	White	func-info-500	Left: func-info-500
Dark Mode	Same values — Toasts are always dark for maximum contrast and focus			

4.19 Calendar / Date Picker

Element	Light Mode	Dark Mode
Header Background	navy-500	navy-800
Header Text	White	slate-0
Weekday Labels	slate-500	slate-300
Day Default	text-primary	text-primary
Day Hover	navy-50	navy-700
Day Selected	green-500 bg, navy-900 text	green-500 bg, navy-900 text
Day Today	navy-500 border, navy-500 text	green-500 border, green-500 text
Day Other Month	slate-300	navy-600
Day Disabled	slate-200 text	navy-700 text
Range Start/End	navy-500 bg, White text	green-500 bg, navy-900 text
Range Between	green-100	green-900 /30%

4.20 Charts & Data Visualization

Token	Light Mode	Dark Mode	Usage
chart-primary	navy-500	navy-400	Primary data series
chart-secondary	navy-300	navy-500	Secondary series
chart-accent	green-500	green-400	Highlight, target, positive
chart-success	green-600	green-500	Positive trend
chart-warning	func-warn-500	func-warn-400	Caution trend
chart-error	func-error-500	func-error-400	Negative trend
chart-grid	slate-200	navy-700	Grid lines
chart-axis	slate-400	slate-500	Axis text
chart-tooltip-bg	navy-900	slate-0	Tooltip background
chart-tooltip-text	White	navy-900	Tooltip text
chart-zero-line	navy-900	slate-0	Zero baseline

Categorical Palette (for multi-series): 1. navy-500 — #1A2B6B 2. green-500 — #99F36C 3. navy-300 — #9FA6C1 4. green-700 — #6EAE52
5. slate-500 — #6B7494 6. green-300 — #D4FAC1 7. navy-700 — #162251 8. slate-300 — #B8BCCC

5. Gradient System (UI Application)

Apply gradients sparingly to hero surfaces, emphasis cards, and energy indicators.

Gradient Name	Direction	Stops	Light Mode Usage	Dark Mode Usage
Energy	135deg	navy-500 → green-500	Hero sections, CTA banners, premium cards	Hero sections, energy indicators
Depth	180deg	navy-900 → navy-500	Dark mode page backgrounds, footer	Atmospheric backgrounds
Glow	135deg	green-300 → green-600	Success states, active indicators, badges	Active glows, hover highlights
Dusk	160deg	navy-800 → #3D2B6B	Special feature cards, dark mode emphasis	Night-mode aesthetics
Surface Shine	180deg	slate-0 → slate-50	Light mode elevated cards	—

Gradient Text (for headlines): - Use Energy gradient with background-clip: text for display headings. - Light mode: navy-500 to green-500. - Dark mode: green-400 to green-500 for luminous effect.

6. Shadow & Elevation System

Shadows are color-aware. They inherit the ambient hue of their environment.

6.1 Light Mode Shadows (Navy-tinted)

Token	Value	Usage
shadow-xs	0 1px 2px rgba(26,43,107,0.05)	Subtle borders replacement
shadow-sm	0 1px 3px rgba(26,43,107,0.08)	Cards, buttons
shadow-md	0 4px 12px rgba(26,43,107,0.10)	Dropdowns, popovers
shadow-lg	0 8px 24px rgba(26,43,107,0.12)	Modals, drawers
shadow-xl	0 16px 48px rgba(26,43,107,0.15)	Full-screen overlays, toasts
shadow-focus	0 0 0 3px rgba(153,243,108,0.30)	Focus rings (universal)
shadow-focus-error	0 0 0 3px rgba(239,68,68,0.25)	Error focus rings

6.2 Dark Mode Shadows (Neutral/Deep)

Token	Value	Usage
shadow-dark-xs	0 1px 2px rgba(0,0,0,0.30)	Subtle depth
shadow-dark-sm	0 1px 3px rgba(0,0,0,0.40)	Cards, buttons
shadow-dark-md	0 4px 12px rgba(0,0,0,0.50)	Dropdowns
shadow-dark-lg	0 8px 24px rgba(0,0,0,0.60)	Modals
shadow-dark-xl	0 16px 48px rgba(0,0,0,0.70)	Overlays
shadow-focus-dark	0 0 0 3px rgba(153,243,108,0.25)	Dark mode focus

7. Typography Color System

7.1 Headings

Level	Light Mode	Dark Mode	Weight	Usage
H1 Display	text-primary navy-900	text-primary White	800	Page titles, hero text
H2	text-primary navy-900	text-primary White	700	Section headers
H3	text-primary navy-900	text-primary White	600	Card titles, subsections
H4	text-primary navy-900	text-primary White	600	Widget titles
H5	text-secondary slate-500	text-secondary slate-300	600	Labels, small headers
H6	text-tertiary slate-400	text-tertiary slate-500	600	Overlines, captions

7.2 Body & Paragraphs

Style	Light Mode	Dark Mode	Size	Usage
Body Large	text-primary	text-primary	18px	Lead paragraphs
Body Default	text-primary	text-primary	16px	Standard text
Body Small	text-secondary	text-secondary	14px	Descriptions
Caption	text-tertiary	text-tertiary	12px	Metadata, timestamps

7.3 Interactive Text

Element	Light Mode	Dark Mode	State
Link Default	navy-500	green-400	—
Link Hover	green-600	green-300	Underline
Link Visited	navy-700	green-500	—
Link Active	navy-900	green-500	—
Button Text (Primary)	White	navy-900	—
Button Text (Accent)	navy-900	navy-900	—
Nav Active	navy-900 + green-500 indicator	White + green-500 indicator	—

8. Interaction State Patterns

8.1 Hover Elevation

- Cards: shadow-sm → shadow-md + border-primary tint
- Buttons: Background darkens 1 step, shadow-sm appears
- Links: Color shifts to green-600 (light) / green-300 (dark) + underline

8.2 Active / Pressed

- Background darkens 2 steps from default
- Inset shadow: `inset 0 2px 4px rgba(0,0,0,0.10)` (light) / `rgba(0,0,0,0.30)` (dark)
- Transform: `translateY(1px)` for tactile feedback

8.3 Focus Visible (Accessibility)

- Universal: `0 0 0 3px rgba(153,243,108,0.30)` (Green glow)
- Error fields: `0 0 0 3px rgba(239,68,68,0.25)` (Red glow)
- Never rely on color alone for focus indication.

8.4 Disabled States

- Opacity: `0.5` universally
- Background: `slate-100` (light) / `navy-800` (dark)
- Text: `slate-300` (light) / `slate-600` (dark)
- Border: `slate-200` (light) / `navy-700` (dark)
- Cursor: `not-allowed`
- No hover elevation change.

8.5 Loading / Skeleton

- Background: `slate-100` (light) / `navy-800` (dark)
- Shimmer: Linear gradient animation from transparent to `slate-50` / `navy-700` and back.
- Duration: `1.5s` ease-in-out infinite.

9. Accessibility & Contrast Compliance

9.1 WCAG 2.1 AA Validated Pairs

Foreground	Background	Ratio	Result
White	<code>navy-900</code>	17.29:1	✓ AAA
<code>green-500</code>	<code>navy-900</code>	12.68:1	✓ AAA
<code>navy-900</code>	White	17.29:1	✓ AAA
<code>navy-900</code>	<code>green-500</code>	12.68:1	✓ AAA
White	Black	19.32:1	✓ AAA
<code>navy-500</code>	White	9.8:1	✓ AAA
<code>green-600</code>	White	3.2:1	✓ AA Large
<code>navy-900</code>	<code>slate-50</code>	15.4:1	✓ AAA
<code>slate-500</code>	White	5.2:1	✓ AA
<code>slate-400</code>	White	3.1:1	✓ AA Large
White	<code>navy-800</code>	14.1:1	✓ AAA
<code>green-400</code>	<code>navy-900</code>	10.5:1	✓ AAA

9.2 Minimum Contrast Rules

- **Normal text (< 18px):** Minimum 4.5:1 against background.
- **Large text (≥ 18px bold / 24px regular):** Minimum 3:1.
- **UI Components (icons, borders):** Minimum 3:1 against adjacent color.
- **Focus indicators:** Must have 3:1 contrast against unfocused state.

9.3 Color-Blind Safety

- Do not use Green vs. Red as the sole differentiator.

- Pair color with icons, text labels, or patterns.
- Success: Checkmark icon + `green-500`
- Error: X icon + `func-error-500`
- Warning: Triangle icon + `func-warn-500`

10. CSS Variables Implementation (Quick Start)

```
:root {  
  /* Primitive */  
  --navy-50: #F4F4F8; --navy-100: #E4E6ED; --navy-200: #C3C8D9;  
  --navy-300: #9FA6C1; --navy-400: #5F6B97; --navy-500: #1A2B6B;  
  --navy-600: #18275F; --navy-700: #162251; --navy-800: #131E43;  
  --navy-900: #111935;  
  
  --green-50: #FAFEF8; --green-100: #F3FEED; --green-200: #E4FCD9;  
  --green-300: #D4FAC1; --green-400: #B8F798; --green-500: #99F36C;  
  --green-600: #85D360; --green-700: #6EAE52; --green-800: #588A44;  
  --green-900: #416535;  
  
  --slate-0: #FFFFFF; --slate-50: #F6F7F9; --slate-100: #EAEBF0;  
  --slate-200: #D6D8E1; --slate-300: #B8BCCC; --slate-400: #8F95AE;  
  --slate-500: #6B7494; --slate-600: #565D76; --slate-700: #404559;  
  --slate-800: #2D313E;  
  
  /* Semantic – Light (Default) */  
  --bg-page: var(--slate-50);  
  --bg-surface: var(--slate-0);  
  --bg-surface-elevated: var(--slate-0);  
  --bg-primary: var(--navy-500);  
  --bg-primary-hover: var(--navy-600);  
  --bg-primary-active: var(--navy-700);  
  --bg-accent: var(--green-500);  
  --bg-accent-hover: var(--green-600);  
  --bg-accent-active: var(--green-700);  
  --bg-muted: var(--navy-50);  
  
  --text-primary: var(--navy-900);  
  --text-secondary: var(--slate-500);  
  --text-tertiary: var(--slate-400);  
  --text-placeholder: var(--slate-300);  
  --text-inverse: var(--slate-0);  
  --text-on-primary: var(--slate-0);  
  --text-on-accent: var(--navy-900);  
  --text-link: var(--navy-500);  
  --text-link-hover: var(--green-600);  
  
  --border-default: var(--slate-200);  
  --border-subtle: var(--slate-100);  
  --border-strong: var(--slate-300);  
  --border-primary: var(--navy-500);  
  --border-accent: var(--green-500);  
  --border-focus: var(--green-500);  
  
  --icon-primary: var(--navy-900);  
  --icon-secondary: var(--slate-500);  
  --icon-tertiary: var(--slate-400);  
  --icon-on-primary: var(--slate-0);  
  --icon-on-accent: var(--navy-900);  
  --icon-accent: var(--green-500);  
  
  /* Shadows */  
  --shadow-xs: 0 1px 2px rgba(26,43,107,0.05);  
  --shadow-sm: 0 1px 3px rgba(26,43,107,0.08);  
  --shadow-md: 0 4px 12px rgba(26,43,107,0.10);  
  --shadow-lg: 0 8px 24px rgba(26,43,107,0.12);  
  --shadow-xl: 0 16px 48px rgba(26,43,107,0.15);  
  --shadow-focus: 0 0 0 3px rgba(153,243,108,0.30);  
}  
  
[data-theme="dark"] {  
  --bg-page: var(--navy-900);  
  --bg-surface: var(--navy-800);  
  --bg-surface-elevated: var(--navy-700);  
  --bg-primary: var(--navy-400);  
  --bg-primary-hover: var(--navy-300);  
  --bg-primary-active: var(--navy-200);  
  --bg-accent: var(--green-500);  
  --bg-accent-hover: var(--green-400);  
  --bg-accent-active: var(--green-300);  
  --bg-muted: var(--navy-800);  
  
  --text-primary: var(--slate-0);  
  --text-secondary: var(--slate-300);  
  --text-tertiary: var(--slate-500);  
}
```

```
--text-placeholder: var(--slate-600);
--text-inverse: var(--navy-900);
--text-on-primary: var(--navy-900);
--text-on-accent: var(--navy-900);
--text-link: var(--green-400);
--text-link-hover: var(--green-300);

--border-default: var(--navy-700);
--border-subtle: var(--navy-800);
--border-strong: var(--navy-600);
--border-primary: var(--navy-400);
--border-accent: var(--green-500);
--border-focus: var(--green-400);

--icon-primary: var(--slate-0);
--icon-secondary: var(--slate-300);
--icon-tertiary: var(--slate-500);
--icon-on-primary: var(--navy-900);
--icon-on-accent: var(--navy-900);
--icon-accent: var(--green-500);

--shadow-xs: 0 1px 2px rgba(0,0,0,0.30);
--shadow-sm: 0 1px 3px rgba(0,0,0,0.40);
--shadow-md: 0 4px 12px rgba(0,0,0,0.50);
--shadow-lg: 0 8px 24px rgba(0,0,0,0.60);
--shadow-xl: 0 16px 48px rgba(0,0,0,0.70);
--shadow-focus: 0 0 3px rgba(153,243,108,0.25);
}
```

11. Usage Ratio Checklist

When designing any screen, verify: - [] **60%** of the area uses Slate neutrals (backgrounds, cards, dividers). - [] **30%** of the area uses Navy structure (headers, primary buttons, text, borders). - [] **10%** of the area uses Solar Green accent (active states, success, CTAs, highlights). - [] Green never exceeds 10% of the viewport to preserve its energy impact. - [] Gradients are used on no more than 15% of the screen and never on dense text backgrounds.

12. Summary Tables

12.1 Token Quick Reference by Component

Component	Key Background	Key Text	Key Border	Key Accent
Button Primary	bg-primary	text-on-primary	—	—
Button Accent	bg-accent	text-on-accent	—	—
Input	bg-surface	text-primary	border-default	border-focus
Card	bg-surface	text-primary	border-subtle	—
Modal	bg-surface	text-primary	border-subtle	—
Alert Success	bg-success	text-success	border-success	icon-success
Alert Error	bg-error	text-error	border-error	icon-error
Nav Active	bg-muted	text-primary	—	green-500 indicator
Tab Active	—	text-primary	green-500 underline	—
Badge Accent	bg-accent	text-on-accent	—	—
Tooltip	navy-900	White	—	—
Toast	navy-900	White	—	Left accent bar

12.2 Mode Comparison Matrix

Token	Light	Dark	Strategy
Page BG	slate-50	navy-900	Inversion of depth
Surface	White	navy-800	Deep atmosphere
Primary Text	navy-900	White	Maximum contrast
Secondary Text	slate-500	slate-300	Muted but readable
Primary Button	navy-500	navy-400	Lighter in dark for visibility
Accent	green-500	green-500	Immutable — energy signal
Focus Ring	green-500 /30%	green-500 /25%	Consistent accessibility
Link	navy-500	green-400	Green becomes navigational signal in dark
Border	slate-200	navy-700	Structural separation
Shadow	Navy-tinted	Pure black	Dark mode needs stronger depth